

INTERMEDIATE PART-II EXAMINATION (MOCK PAPER)

COMPUTER SCIENCE (12th Class)

Test Syllabus: C-Language (Chapter 8 to 14)

Total Marks: 75

Time Allowed: 2:30 Hours

SECTION A – OBJECTIVE (15 Marks)

Note: Attempt all questions. Each question carries 1 mark. Select the most appropriate option.

1. The output of the compiler is called:
A) Source Code
B) Object Code
C) Unicode
D) Linker Code
2. Which format specifier is used for a single character?
A) %s
B) %d
C) %c
D) %f
3. What will be the output of: `int a = 10; printf("%d", a++);`?
A) 10
B) 11
C) 12
D) Error
4. Which escape sequence is used for a Backspace?
A) \n
B) \b
C) \t
D) \r
5. In `if(x = 5)`, the condition is treated as:
A) False
B) True
C) Error
D) Null
6. `switch` statement is an alternative to:
A) For loop
B) If-else-if
C) While loop
D) Functions
7. What is the output of: `int i=1; while(i<5) { i++; } printf("%d", i);`?
A) 4
B) 5

- C) 6
 - D) Infinite
8. Which loop is best when the number of iterations is known in advance?
- A) while
 - B) do-while
 - C) for
 - D) All of them
9. A variable declared inside a function is called:
- A) Global Variable
 - B) Static Variable
 - C) Local Variable
 - D) External Variable
10. The default return type of a function in C is:
- A) void
 - B) int
 - C) char
 - D) float
11. fprintf is used for:
- A) Formatted console input
 - B) Formatted file output
 - C) String handling
 - D) Memory allocation
12. Which operator has the lowest precedence?
- A) =
 - B) +
 - C) *
 - D) &&
13. What is the result of $10 \% 3$?
- A) 3.33
 - B) 3
 - C) 1
 - D) 0
14. To close a file, we use the function:
- A) exit()
 - B) fclose()
 - C) close()
 - D) terminate()
15. The logical NOT operator is represented by:
- A) !
 - B) !=
 - C) ~
 - D) <>

SECTION B – SHORT QUESTIONS (45 Marks)

Note: The section is divided into three parts. Attempt any 6 questions from each part. Each question carries 2.5 marks.

Part I

1. Identify the error in the following code:
`void main() { int x = 10; printf("Value = %d", &x); }`
2. What will be printed by the following code?
`int a = 5, b = 10;
printf("%d", a + b * 2);`
3. Write a C statement to print the value of PI (3.14159) using a define directive.
4. Trace the error in this variable declaration:
`int 1st_Number; float @price;`
5. What is the difference between `getch()` and `getche()`?
6. Define Delimiters. Give an example.
7. Differentiate between a Unary Operator and a Binary Operator.
8. Determine the output of:
`printf("Pakistan\nZindabad");`
9. Why is `#include <stdio.h>` used in C programs?

Part II

1. What is the output of the following segment?
`int x = 5;
if (x == 5) printf("Equal");
else printf("Not Equal");`
2. Find the error in the following code:
`int k = 0;
while (k < 5)
printf("%d", k);
k++;`
3. Convert the following for loop into a while loop:
`for(int i=1; i<=10; i++) { printf("%d", i); }`
4. Predict the output of the following code:
`int c = 0;
switch(c) {
case 0: printf("Zero");
case 1: printf("One"); break;
}`
5. What is a Nested Loop? Give the syntax.
6. Differentiate between `break` and `continue` statements.
7. Write a condition using the logical AND (&&) operator to check if a variable `n` is between 10 and 20.
8. What is a Sentinel Controlled Loop?

9. Define Control Structure. Name its types.

Part III

1. Predict the output:

```
void func() {  
    int x = 10;  
    x++;  
    printf("%d", x);  
}
```

// Assume func is called twice. What is printed?

2. Find the error in the following function:
`void sum(int a, int b) { return a + b; }`
3. What does `fopen("data.txt", "w")`; do if the file "data.txt" already exists?
4. Differentiate between Local and Global variables.
5. What is a Function Prototype? Why is it required?
6. Explain the purpose of the EOF marker.
7. Write the syntax of a function definition.
8. What is the difference between `printf()` and `fprintf()`?
9. Define Actual Parameters and Formal Parameters.

SECTION C – LONG QUESTIONS (15 Marks)

Note: Attempt any 3 questions. Each question carries 5 marks.

1. Write a program in C that inputs a character from the user and checks whether it is a Vowel or a Consonant using the switch statement.
2. Write a program in C that inputs a number and checks whether it is a Prime Number or Composite Number using a loop.
3. Write a program that accepts a number from the user and prints its Table (multiplication table) up to 10 using a for loop.
4. Explain the if-else-if statement. Describe its syntax and working with the help of a flowchart.
5. Define Function. Explain the difference between Call by Value and Call by Reference with examples.